

Freshness & Defensibility Scan: De Novo Computationally-Designed Urethanase (as of June 1, 2026)

TL;DR

- **The flag is still open.** As of June 1, 2026, there is no publicly disclosed, truly de novo computationally-designed urethanase/carbamate hydrolase — no preprint, no peer-reviewed paper, no patent, no conference abstract, no thesis chapter, no company announcement. Every "urethanase" hit in 2023–2026 is either (a) metagenomic mining or (b) engineering/redesign of a pre-existing natural scaffold (UMG-SP1/2/3 family, AbPURase, Aes72, GatA250, etc.). The Baker Lab and competitors have published de novo designs for esters (Lauko, Science Feb 2025), zinc phosphonate ester hydrolysis (Kim/Woodbury/Ahern, Nature Dec 3 2025), cysteine hydrolases (RFdiffusion3 bioRxiv Sept 2025), and peptide cysteine proteases (bioRxiv Nov 21 2025) — but never the urethane $C(=O)-NH$ bond.
- **The enabling toolchain is fully in place and the field velocity is very high.** RFdiffusion2 (Nat Methods 2025) and RFdiffusion3 (bioRxiv Sept 2025) can scaffold arbitrary theozymes from atom-level geometries; Lauko's PLACER ensemble method directly solves the multi-step Ser-His-substrate problem that a urethanase needs; the Baker lab's Kiera Sumida is publicly named (Baker Lab announcement, Feb 13, 2025) as "already applying these methods to tackle plastic degradation"; and the Ramos/Fernandes/Bornscheuer/Wei consortium has published high-resolution QM/MM mechanism + transition-state geometries for UMG-SP2 (Paiva et al., Chem. Sci. 2025, 16:2437–2452, overall Gibbs activation energy $20.8 \text{ kcal}\cdot\text{mol}^{-1}$, Stage 1 acylation rate-limiting) and UMG-SP1 (Świderek et al., JACS 2025, doi:10.1021/jacs.5c13147, calculated $21.2 \text{ kcal}\cdot\text{mol}^{-1}$ vs. $22.9 \text{ kcal}\cdot\text{mol}^{-1}$ experimental TOF) that could be dropped directly into RFdiffusion2 as a theozyme. The probability that a de novo urethanase preprint drops in 2026 is high, but as of June 1, 2026 it has not. (Nature + 4)
- **The team's specific angle — de novo urethanase for elastane/spandex in children's apparel — is the cleanest remaining whitespace.** No de novo design work has been publicly framed around textile/elastane (vs. PU foam). The closest adjacent claim is the LYCRA Company × Samsara Eco partnership: Samsara's Jerrabomberra plant opened September 3, 2025, and CEO Paul Riley stated, "In just four years, we've scaled from bench research through to pilot, demonstration, and now our first plant. This is a true tipping point for circularity." But Samsara's platform (EosEco) is publicly described as "AI-crafted enzymes" via a proprietary "AI-powered enzyme discovery and development platform" — i.e., AI-guided screening/evolution of natural enzymes, NOT from-scratch generative design. The team can credibly claim the de novo + elastane intersection as untaken — but should plant the flag at the hackathon (June 14–17, 2026) immediately, because the window of clean novelty is plausibly weeks, not months. (Textileworld + 3)

Key Findings

Category 1 — Truly DE NOVO Designed Urethanases / Carbamate Hydrolases (THE KEY CATEGORY)

Status: ZERO public disclosures meeting the strict definition, as of June 1, 2026.

After exhaustive search of bioRxiv, ChemRxiv, arXiv, Nature/Science portfolios, PNAS, JACS, ACS Catalysis, Angewandte, Green Chem, ACS SusChemEng, Protein Science, Protein Engineering Design and Selection, Google Patents, USPTO image-ppubs, WIPO, the Baker Lab and IPD news pages, GeekWire, C&EN, the ACS Spring 2026 (San Diego, March 2026) program, Gordon/RSC/EMBO conference listings, and company technical disclosures from Samsara Eco, Carbios, Epoch Biodesign, Profluent, Cradle.bio, Protein Evolution Inc., and Liberation Labs, no preprint, paper, patent, abstract, or thesis discloses an enzyme whose backbone and active site were generated computationally from scratch (RFdiffusion-family, Rosetta theozyme matching/inside-out, GENzyme, ESM-based generative, ProGen, or any other generative approach) and experimentally shown to hydrolyze a urethane/carbamate bond.

Every "computational" urethanase paper in the period is **redesign of an existing natural scaffold**, not from-scratch design:

- Zheng/Liu et al., "Computational Redesign of a Urethanase for Efficient Polyurethane Depolymerization," ACS Catalysis 2026, doi:10.1021/acscatal.5c08984 — QM/MM-guided point mutations on UMG-SP1. Best variant L126M, ~6× improvement. NOT de novo.

ACS Publications
- Świderek et al., "Insights into the Catalytic Activity of a Metagenome-Derived Urethanase," JACS 2025, 147(46):42511–42523, doi:10.1021/jacs.5c13147 — QM/MM mechanism of UMG-SP2. Mechanistic, not a design paper.
- Paiva et al., "Unveiling the enzymatic pathway of UMG-SP2 urethanase," Chemical Science 2025, 16:2437–2452, doi:10.1039/D4SC06688J — QM/MM atomic-level mechanism. Critically, this paper provides the kind of transition-state geometry that a de novo designer would need as a theozyme.

The absence-of-evidence caveat: Industrial labs (Samsara Eco, Epoch Biodesign, Profluent, Cradle.bio, Carbios, the LYCRA-Samsara collaboration, the Baker Lab's spin-out Xaira, Liberation Labs) may have unpublished/confidential work that crosses this line. But based on press releases, Fashion-for-Good profiles, technical blog posts, and patent filings up to June 1, 2026, none of these companies publicly frames its work as "de novo generative design of a from-scratch protein" — they all describe directed evolution, ML-guided variant selection, or sequence mining of natural enzyme libraries (Samsara: "AI-crafted enzymes" via EosEco discovery/development platform; Epoch: AI-screening of enzyme databases plus directed

evolution; Carbios: engineered cutinases). There is no patent on Google Patents/USPTO/WIPO claiming a de novo computationally designed urethanase as of the search date. [Textileworld + 3](#)

Category 2 — ENGINEERED / EVOLVED Natural Urethanases (NOT de novo, catalog only)

Catalogued state-of-the-art; clearly distinct from the team's flag.

- **UMG-SP1/SP2/SP3** (Branson et al., Angew. Chem. Int. Ed. 2023, doi:10.1002/anie.202216220) — original metagenome-mined urethanases for polyether-PU. [Wiley Online Library](#)
- **UMG-SP2 structure-guided engineering**: Li, Han, Cong et al., "Structure-Guided Engineering of a Versatile Urethanase Improves Its Polyurethane Depolymerization Activity," Advanced Science 2025, 12(13):e2416019, doi:10.1002/advs.202416019. Variants A141G (>30.7-fold improvement on polyester-PU particles), Q399A (>7.4-fold more MDA released), D226A — but combining mutations did not stack. [nih](#) [Wiley Online Library](#)
- **UMG-SP1 computational redesign** (Zheng et al., ACS Catal 2026, doi:10.1021/acscatal.5c08984) — best variant L126M $\approx 6\times$ improvement.
- **Aes72 mechanism-guided engineering** for PU depolymerization, Engineering 2026, doi:10.1016/j.eng.2026.02.008.
- **Pastor/Reisky et al.** (RWTH/Covestro), "Identification of Urethanases for Biocatalytic Recycling of TDI/MDI-Based Polyurethanes," ChemSusChem 2025 (PMC12456375) — TflABH, MthABH, OspAmd identified and biochemically characterized. [nih](#)
- **GatA250 amidase** (Xin et al.) shown to cleave urethane bonds in polyester-PU at low efficiency. [PubMed Central](#)
- **Bittner et al., EP3587570A1** (2020, Covestro) — original "novel urethanases" patent on the UMG-SP class.

Category 3 — DISCOVERED / MINED Urethanases (NOT de novo, catalog only)

- **AbPURase / GRASE** (Chen, Sun, Cui, Wu et al.), "Glycolysis-compatible urethanases for polyurethane recycling," Science 2025, 390(6772):503–509, doi:10.1126/science.adw4487 (published Oct 30, 2025). GRASE = Graph-Neural-Network-based Recommendation of Active and Stable Enzymes; $\sim 98.6\%$ PU foam depolymerization in 8 h at kg scale in 6 M diethylene glycol. **Two orders of magnitude greater activity than previously known urethanases.** This is the current SOTA — but it is explicitly mined from databases via GNN, not de novo. This is the single most relevant prior-art bullet the team needs to address: the "AI + urethanase" headline space is publicly taken by Chen/Sun/Wu, but the de novo-from-scratch interpretation of "AI" is not. (NB: The popular-press claim that this is a "Pythia-Pocket / Pythia"-designed enzyme — e.g., warpnews.org, pu-magazine.com, ars technica — is misleading; the underlying Science paper is GRASE-based sequence selection, not from-scratch backbone design.) [Science](#) [Phys.org](#)

- **Metagenomic drain urethanases for elastane** — Anselmi S, Ni Y, Tonoli A, Wu J, Wang Y, Prout L, Miodownik M, Jeffries JWE, Hailes HC (UCL Dept. of Chemistry / UCL Plastic Waste Innovation Hub), "The discovery of new metagenomic urethanases utilising a novel colorimetric assay for applications in the biodegradation of polyurethanes," Green Chem., doi:10.1039/d5gc03560k, published September 4, 2025 (PMC12421990). Three novel urethane hydrolases from a drain metagenome shown to break down MDA-based elastane model substrates and commercial fabrics. This is the most directly competitive prior art on the **elastane specifically** angle — it claims metagenomic discovery on elastane substrates. The team's "de novo + elastane" angle survives because this is mining, not design.
- **u10/u15/u17 bacterial urethanases** — Li Z, Han X, Cong L et al. (same Bornscheuer/Wei/Liu consortium), "Expanding the Enzymatic Landscape for Polyurethane Degradation of Novel Bacterial Urethanases," bioRxiv doi:10.64898/2026.02.11.705263, posted February 11, 2026. Six new natural urethanases mined bioinformatically; u15 emerged as particularly notable for its catalytic efficiency on DUE-MDA, with a k_{cat}/K_M of $51.8 \pm 0.1 \text{ s}^{-1} \cdot \text{mM}^{-1}$. Mining, not design.

Category 4 — DE NOVO Designed Hydrolases for OTHER Bond Types (the enabling toolchain)

This is where the field velocity story is most clearly visible. Every adjacent bond type has been claimed in the last ~18 months, and the toolchain is now "shovel-ready" for carbamates.

Work	Substrate / Bond	Date	Method	Result	Topic
Holst et al., "De novo design of a polycarbonate hydrolase," Protein Eng Des Sel 2023, 36:gzad022, doi:10.1093/protein/gzad022 (bioRxiv 2023.03.10.532063)	Carbonate diester (in polycarbonate)	Mar 2023 / pub 2023	Rosetta theozyme matching, thermostable scaffolds	Activity confirmed by AFM on Makrolon 2808; Tm > 95 °C. No kinetics reported.	Hydrolysis of polycarbonates
Lauko, Pellock, Sumida et al., "Computational design of serine hydrolases," Science 2025, 388:eadu2454, doi:10.1126/science.adu2454	Ester (4-MU acetate; multistep acylation/deacylation)	Feb 13 2025	RFdiffusion + PLACER ensemble preorganization	kcat/Km up to 2.2×10 ⁵ M ⁻¹ s ⁻¹ ; five novel folds; Bakerlab rate enhancement >10 ⁸	Various topics

Work	Substrate / Bond	Date	Method	Result	Topic
Ahern, Yim, Tischer, Salike, Woodbury, Kim et al., "Atom-level enzyme active site scaffolding using RFdiffusion2," Nat. Methods 2025, doi:10.1038/s41592-025-02975-x (bioRxiv 2025.04.09.648075)	Retroaldolase + cysteine hydrolase + zinc hydrolase (esterase)	Apr 10 2025 preprint / Nat Methods 2025	RFdiffusion2 atom-level theozyme conditioning	Active enzymes from <96 designs in each of 3 reactions (bioRxiv Tamarind)	Various
Kim, Woodbury, Ahern et al., "Computational design of metallohydrolases," Nature 2025/2026 (online Dec 3, 2025), doi:10.1038/s41586-025-09746-w	Zinc-catalyzed phosphonate-ester hydrolysis (4-MU-PA)	Dec 3, 2025	RFdiffusion2 + QM theozyme + GLU general base	kcat/KM 53,000 M ⁻¹ s ⁻¹ ; kcat 1.5 s ⁻¹ — orders of magnitude above prior designed metallohydrolases	Hydrolysis
Butcher, Krishna et al., "De novo Design of All-atom Biomolecular Interactions with RFdiffusion3," bioRxiv 2025.09.18.676967 (Sept 18 / updated Nov 19, 2025)	Cysteine hydrolases (and DNA-binders)	Sept–Nov 2025	RFdiffusion3 all-atom diffusion conditioning on ligand atoms	DNA binders + cysteine hydrolases; 1/10 compute cost vs. RFdiffusion2	Various
Baker Lab "Computational design of cysteine proteases" (Lauko, Sumida et al.), bioRxiv 2025.11.21.689808	Amide bonds in peptides (Dokki series)	Nov 21, 2025	RFdiffusion + PLACER, Cys-His-Asp triad	Crystal structures matching design at 3.38 Å; active enzymes	Proteases

Work	Substrate / Bond	Date	Method	Result	T tc
Hua, Lu et al., "GENzyme: reaction-conditioned de novo enzyme design," arXiv:2411.16694 (2024-2025)	Reaction-conditioned, multiple EC classes	Nov 2024	Three-stage generative (pocket gen + inpaint + binding screen) arxiv	In silico benchmarks across EC classes	M sc av ca hy de es

Kiera Sumida (Baker Lab PhD trainee) is publicly named (Baker Lab announcement, Feb 13, 2025) as "already applying these methods to tackle plastic degradation." Her only first-author preprint of record is Sumida et al., bioRxiv 2023.10.03.560713 (ProteinMPNN-based improvement of myoglobin & TEV protease, not a urethanase). UW Chem Graduate Program Excellence Award (June 17, 2025) describes her PhD topic as "de novo protein design tools to create novel plastic-degrading enzymes" without naming a substrate. No urethanase preprint from her has appeared by June 1, 2026 — but she is the single most plausible name to watch.

bioRxiv + 2

Field Velocity & Imminence Assessment

- The chain Lauko (Feb 2025, esters) → Ahern/RFdiffusion2 (Apr 2025, Z-hydrolase + Cys hydrolase + retroaldol) → Butcher/RFdiffusion3 (Sept 2025, all-atom ligand conditioning) → Kim/Woodbury (Dec 3 2025, Zn metallohydrolases at near-native efficiency) → Baker cysteine protease (Nov 21 2025, peptide amides) → **(missing link: carbamate)** is so tight that the team should assume a competing de novo carbamatase preprint could appear between June 1 and June 17, 2026, but there is no specific announced or in-press signal that one will.
- Specific signals to monitor weekly between now and the hackathon (June 14-17, 2026):
 1. New bioRxiv posts from Baker, Lauko, Sumida, Pellock, Ahern, Woodbury, Kim, Kalvet — particularly with "carbamate", "urethanase", "polyurethane", or "elastane" in the title.
 2. The Bornscheuer / Wei / Ramos / Fernandes consortium (they have the mechanism in hand and a strong PU portfolio).
 3. Samsara Eco's LYCRA-collaboration lab in Jerrabomberra (opened September 3, 2025) — but historically they do not preprint.
 4. ACS Sustainable Chem & Eng / Green Chem April-June 2026 issues.
- No announced talk, in-press paper, or scheduled press release in the searched signals points to an imminent de novo urethanase publication. The probability of being scooped between now and June 17, 2026 is non-trivial but not high.

Specific Whitespace for the Cox/Carter's Hackathon Team

The clean intersection that remains unclaimed as of June 1, 2026:

1. **De novo backbone (RFdiffusion2 or RFdiffusion3) + carbamate theozyme** — open.
2. **Targeted at elastane/spandex/PU specifically** — open as a de novo project.
3. **Motivated by textile (vs. foam) circularity** — open (Samsara/LYCRA collaboration exists but is not de novo).
4. **Children's apparel use case (small, blended, mixed-fiber, low-feedstock-volume)** — completely open; no published or company work on this micro-vertical.

The single biggest "you are very close to being scooped" risk is that the Chen/AbPURase/GRASE work is already publicly framed in the press as "AI-designed enzyme for polyurethane" (which is misleading — it's mining), so a non-specialist audience may already perceive the flag as planted. **The team needs to be explicit and pedantic about the distinction between mined-via-AI (GRASE/AbPURase) and computationally designed de novo (their proposed work) in every piece of project communication.**

Details

What "de novo computationally designed" must mean for the flag to be defensible

The crisp definition the team should use, and the one this scan was conducted against:

- Backbone generated from scratch via a generative model (RFdiffusion-N, Chroma, Genie, FrameDiff, FoldFlow, RFdiffusion3 all-atom, GENzyme, etc.) or via Rosetta theozyme + inside-out matching to non-homologous scaffolds.
- Active-site geometry specified explicitly (theozyme) for the cleavage of the carbamate -O-C(=O)-NH- bond, NOT borrowed wholesale from a natural urethanase/amidase/esterase scaffold.
- Sequence designed (LigandMPNN, ProteinMPNN, Rosetta) on the generated backbone.
- Experimentally expressed and shown to cleave a carbamate substrate (model dicarbamate, BBC, DUE-MDA, elastane model, etc.).

Engineering UMG-SP, AbPURase, or GatA250 by mutation — even ML-guided mutation — does not qualify. Neither does GNN-based selection of natural enzymes (GRASE).

Why a serine-hydrolase-style de novo urethanase is mechanistically tractable

The carbamate hydrolysis transition state is geometrically very close to that of an ester or peptide amide: a Ser-O⁻ nucleophilic attack on the carbamate carbonyl, oxyanion-hole stabilization of the developing negative charge on the carbonyl oxygen, His-mediated proton

shuttling, and deacylation by an activated water — exactly the Lauko architecture. The Ramos/Fernandes/Świderek QM/MM work on UMG-SP1/SP2 has already produced atomic-resolution descriptions of this transition state — Paiva et al., Chem. Sci. 2025, 16:2437–2452 reports an "overall Gibbs activation energy of 20.8 kcal mol⁻¹" with Stage 1 acylation as the rate-limiting step; Świderek et al., JACS 2025, doi:10.1021/jacs.5c13147 reports a calculated free energy barrier of 21.2 kcal·mol⁻¹ vs. 22.9 kcal·mol⁻¹ from experimental TOF. Those geometries can be lifted directly into RFDiffusion2 as a theozyme. This is what makes the Sumida-type project so obviously feasible — and what makes the team's flag genuinely time-sensitive. (RSC Publishing + 2)

Closely adjacent claims to flag in the team's own write-up (for defensibility)

The team should pre-emptively cite and distinguish from these to make their novelty claim robust to skeptical reviewers/judges:

- Holst et al., 2023 polycarbonate hydrolase (different bond, -O-C(=O)-O-).
- Chen et al., AbPURase Science 2025 (mined, not designed).
- Branson 2023 / Li 2025 / Zheng 2026 (engineering UMG-SP).
- Lauko 2025 / Kim 2025 / Butcher 2025 / Ahern 2025 (de novo, different substrates).
- Anselmi/Hailes et al., Green Chem 2025, doi:10.1039/d5gc03560k (metagenomic urethanases on elastane — the closest prior art on the elastane axis, but mining not design).
- Samsara Eco × LYCRA (Sept 3, 2025 plant opening) (commercial, "AI-crafted enzymes" via EosEco discovery/development platform — not from-scratch design).
- Epoch Biodesign / Carbios (nylon 6,6, PET — different polymers).

Companies that are most likely to publish first, if anyone does, before June 17, 2026

- Baker Lab / IPD (Sumida watch).
- Bornscheuer/Wei/Liu (Greifswald/Tianjin) + Ramos/Fernandes (Porto) consortium — they have the mechanism, the scaffolds, and the substrates already; if they pivot to de novo design they would publish quickly.
- Samsara Eco — if they pivot from evolution to design and disclose it. Their public framing remains "AI-crafted enzymes" via EosEco discovery/development platform, but Hitachi Ventures' Jan Marchewski has confirmed the Jerrabomberra hub will "test AI-discovered enzymes at scale" — language compatible with either mining or design.
- Less likely: Profluent (focus is broader protein generation, no PU portfolio publicly), Cradle.bio (no PU portfolio publicly), Carbios (focus is PET, working with engineered cutinases).

Recommendations

1. **Plant the flag now and define it pedantically.** In every piece of hackathon communication

— README, slide deck, Q&A — define "de novo computationally designed urethanase" precisely (backbone-generated + theozyme-specified + sequence-designed + experimentally validated on a carbamate substrate), and explicitly cite-and-distinguish from GRASE/AbPURase (mining), UMG-SP variants (engineering), and the Holst polycarbonate hydrolase (wrong bond). This is the single highest-leverage defensibility move.

2. **Monitor bioRxiv daily, weekly through June 17, 2026.** Set automated alerts on bioRxiv RSS for terms: "urethanase", "carbamate hydrolase", "carbamatase", "polyurethane enzyme", "elastane enzyme", "RFdiffusion polyurethane", and authors {Sumida, Lauko, Pellock, Ahern, Woodbury, Kim D, Kalvet, Baker D, Bornscheuer, Wei R, Liu W, Ramos MJ, Fernandes PA}. Also monitor IPD news (ipd.uw.edu) and Baker Lab pubs page weekly.
3. **Anchor the design on the Paiva/Świderek transition-state geometry** (Chem. Sci. 2025 16:2437–2452; JACS 2025 doi:10.1021/jacs.5c13147), not on UMG-SP sequence. This is the single biggest technical lever: it lets the team honestly describe their work as "we used the QM/MM-derived theozyme from the literature and built a brand-new backbone around it," which is unambiguously de novo even to a hostile reviewer.
4. **Pick elastane (MDA-based dicarbamate model substrate) over PU foam as the demonstration substrate.** This is the cleanest whitespace in both novelty (no de novo work has framed itself around elastane) and in fit-to-track (children's apparel circularity). The Anselmi/Hailes drain-metagenome elastane paper (Green Chem 2025, doi:10.1039/d5gc03560k) gives a validated colorimetric tyrosinase assay that the team can adopt directly; the Branson DUE-MDA assay is also published.
5. **Pre-register the project publicly (even just a one-pager on GitHub or a dated tweet) before June 14, 2026.** This is cheap, takes minutes, and gives the team a timestamped public claim of priority that survives even a competing preprint dropping mid-hackathon.
6. **Benchmarks/thresholds that should change the strategy:**
 - If a Baker Lab / Sumida bioRxiv preprint on a de novo urethanase appears before June 17 → pivot the angle to "first de novo urethanase optimized for elastane / textile substrates / mixed-fiber children's apparel" (their flag was foam; yours is textile).
 - If a competing de novo design paper claims elastane specifically → pivot to "first de novo urethanase + economic LCA for children's apparel circularity" (apply, don't just discover).
 - If GRASE/AbPURase v2 from Chen et al. appears with a de novo backbone claim → the flag is largely gone; pivot to a different bond (e.g., a de novo polyester-PU ester hydrolase that complements an existing urethanase, since both bonds need to be cleaved for full polymer breakdown).
7. **Do not over-promise wet-lab outcomes in a hackathon timeframe.** The Lauko/Kim pipeline still required 96–300+ designs tested experimentally. A hackathon flag is best planted with an in-silico-validated design pipeline + RFdiffusion2 outputs +

AlphaFold2/PLACER preorganization metrics + a designed sequence ready for synthesis, NOT with kinetic data that cannot realistically be produced in three days.

Caveats

- **Absence of evidence vs. evidence of absence.** This scan is exhaustive across public sources but cannot exclude confidential industrial R&D at Samsara Eco, Epoch Biodesign, Profluent, Cradle.bio, Carbios, Protein Evolution Inc., Liberation Labs, the IPD's commercial spin-outs (notably Xaira), or large fashion-industry consortia. The Samsara × LYCRA Jerrabomberra lab — opened September 3, 2025, with CEO Paul Riley calling it "a true tipping point for circularity" — is a specific known unknown. (Specialty Fabrics Review)
- **The "AI-designed enzyme" press-headline space is already publicly occupied** by GRASE/AbPURase (Chen et al., Science Oct 2025). Non-specialist judges and reporters may believe the flag has been planted. The team's defensibility depends entirely on enforcing the technical distinction between "AI-mined natural enzyme" and "AI-designed de novo enzyme."
- **The most direct competitor publication, if one drops in the next 2–6 weeks, is most likely to come from the Baker Lab (Sumida)** with a paper structurally analogous to Lauko 2025 but with a carbamate theozyme. This would be a near-complete planting of the flag. The team should have a contingency pivot ready (Recommendation 6).
- **The "children's apparel" framing is narrative, not technical** — the underlying enzymology is identical to elastane recycling for any apparel category. Defensibility ultimately rests on the de novo + carbamate technical claim, not on the apparel-category framing.
- **The polycarbonate-hydrolase work (Holst 2023) is the closest possible scientific precedent** and the team's most likely "but isn't this basically already done?" critique. The crisp rebuttal: polycarbonate has an -O-C(=O)-O- bond (no nitrogen); polyurethane/elastane has -O-C(=O)-NH- ; the chemistry, theozyme, transition state, and mechanism differ in the proton-donor/acceptor profile and oxyanion-hole geometry. They are mechanistically siblings, not the same.
- **Final epistemic note:** The phrase "as of June 1, 2026" is doing real work in every claim above. The flag is open today. There is no public signal that it will still be open on June 17.